

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

CELL, The Unit of Life

LECTURE-6

ZOOLOGY

DAVOOD SIR (CEO of NG)



Topics To Be Covered

- 1 Ribosome and its types
- 2 Centrosome and Centrioles
- 3 Previous Year NEET Questions.

Eukaryotic Cell



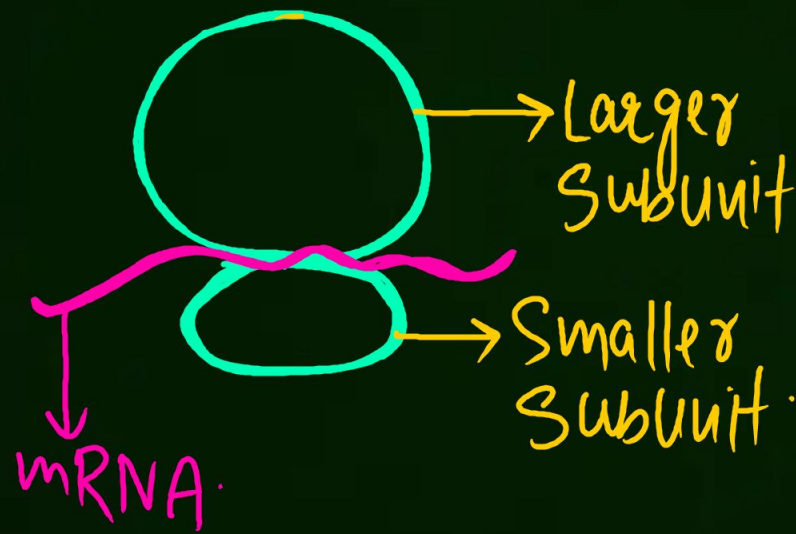
Ribosomes

- Smallest cell organelle with no membrane
- Protein factory of the cell.
- Also known as palade particles

• Discovery: → George Emil Palade (1953).

• Composition: → Protein + tRNA

• Structure:



Types of Ribosome:

80'S Ribosome

70'S Ribosome

S = Svedberg unit → Sedimentation coefficient.

Prokaryotic Ribosome

60'S

50'S

40'S

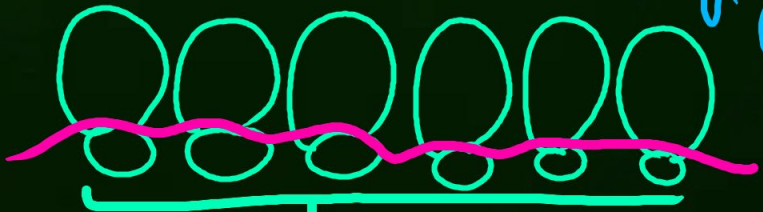
30'S

Eukaryotic ribosome

Made up of 80 types of proteins.

Functions:

→ Ribosome reads mRNA & then synthesizes proteins accordingly.



Poly some.

Ribosomes

- Ribosomes are the membraneless granular structures first observed under the electron microscope as dense particles by **George Palade (1953)**.
- They are composed of **ribonucleic acid (RNA)** and **proteins**.
- The **eukaryotic ribosomes are 80S** while the **prokaryotic ribosomes are 70S**.
- Each ribosome has two subunits, larger and smaller subunits.
- The two subunits of **80S ribosomes are 60S and 40S** while that of **70S ribosomes are 50S and 30S**. Here 'S' (Svedberg's Unit) stands for the sedimentation coefficient.
- Sedimentation coefficient is indirectly a measure of density and size.
- Several ribosomes may attach to a single mRNA and form a chain called **Polyribosomes or polysome**. The Ribosoms of a polysome translate the mRNA into proteins.

Question: NEET-2025

4-Marks



From the statements given below choose the correct option:

- A. The eukaryotic ribosomes are 80S and prokaryotic ribosomes are 70S.
- B. Each ribosome has two sub-units.
- C. The two sub-units of 80S ribosome are 60S and 40S while that of 70S are 50S and 30S.
- D. The two sub-units of 80S ribosome are 60S and 20S and that of 70S are 50S and 20S.
- E. The two sub-units of 80S are 60S and 30S and that of 70S are 50S and 30S.

1

A, B, C are true

2

A, B, D are true

3

A, B, E are true

4

B, D, E are true

Question: NEET-2023*4-Marks*

Eukaryotic
How many different proteins does the ribosome consist of?

1

40

2

20

3

80

4

100



Question: NEET-2019



4-Mark 9.

Match List-I with List-II.

Column-I

- A. Golgi apparatus
- B. Lysosomes
- C. Vacuoles
- D. Ribosomes

Column-II

- I. Synthesis of protein
- II. Trap waste and excretory products
- III. Formation of glycoproteins and glycolipids
- IV. Digesting biomolecules

Choose the correct answer from the options given below

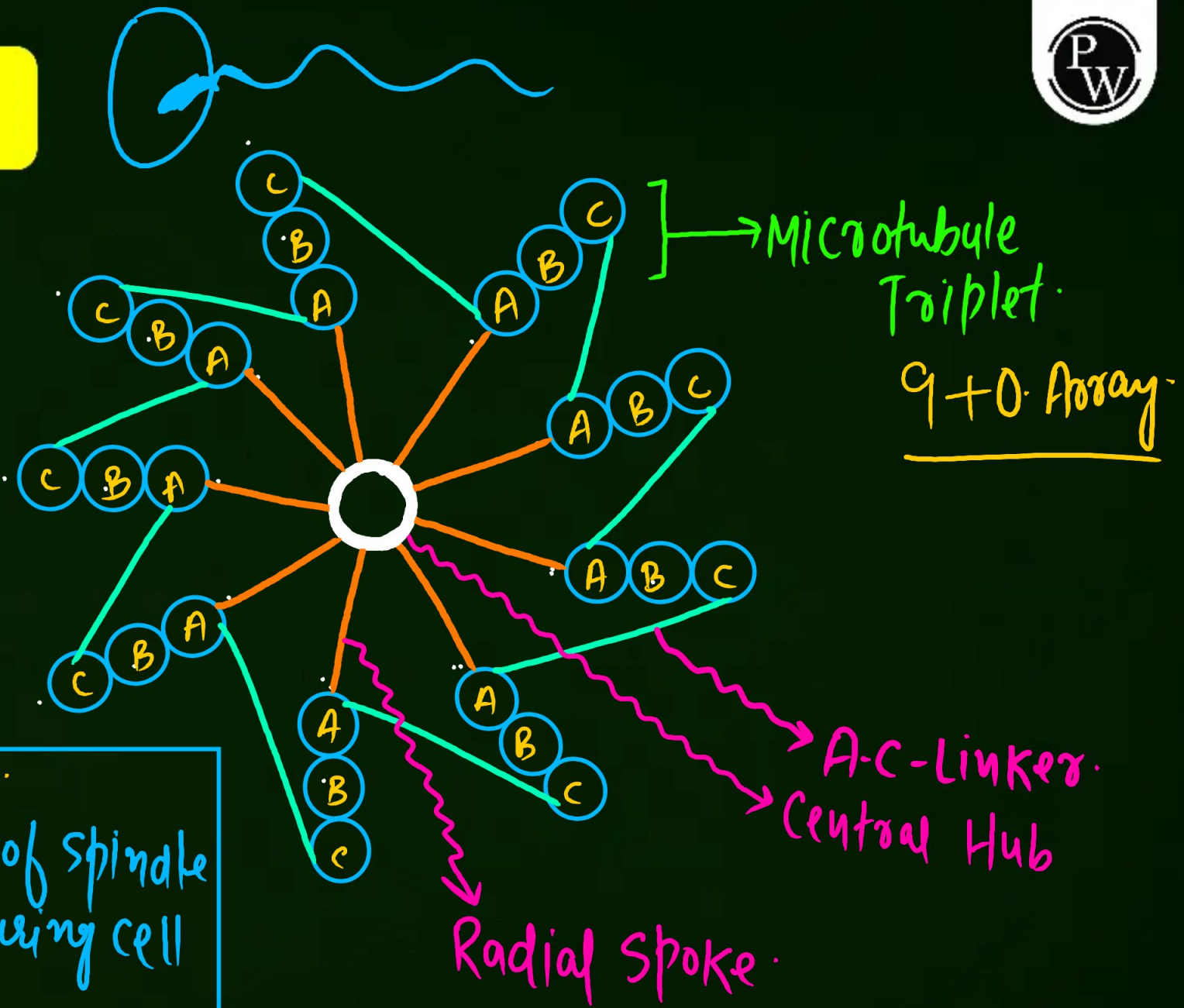
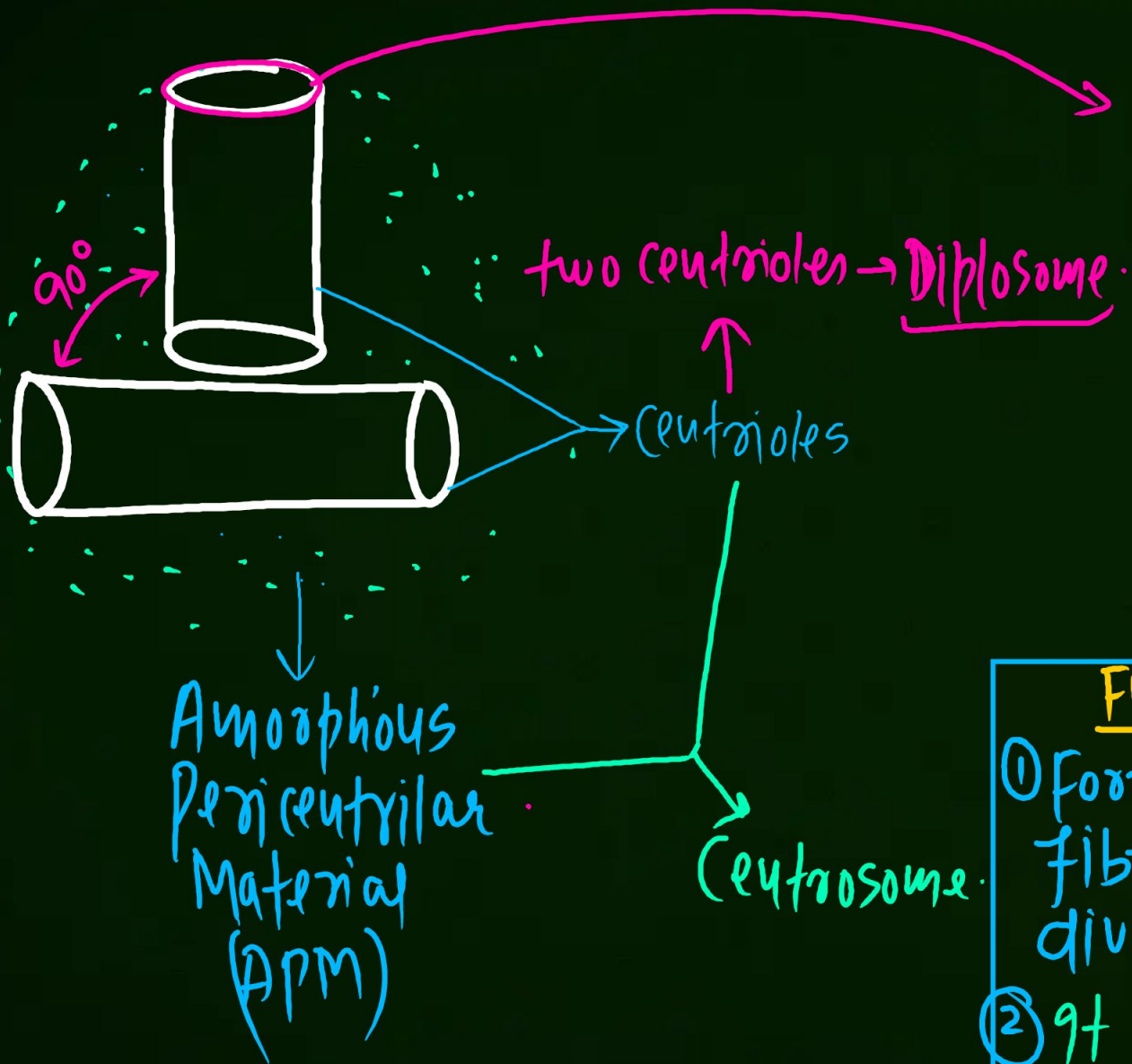
1 A-(iii), B-(iv), C-(ii), D-(i)

2 A-(iv), B-(iii), C-(i), D-(ii)

3 A-(iii), B-(ii), C-(iv), D-(i)

4 A-(i), B-(ii), C-(iv), D-(iii)

Centrosome and Centrioles



- Function:
- ① Formation of spindle fibres during cell division.
 - ② 9+ forms basal body of flagella.

Centriole Byya Cartwheel bangaya



Centrosome and Centrioles

Centrosome is an organelle usually containing two cylindrical structures called **centrioles**. They are surrounded by **amorphous pericentriolar materials**. Both the centrioles in a centrosome lie perpendicular to each other in which each has an organisation like the **cartwheel**. They are made up of nine evenly spaced peripheral **fibrils** of **tubulin protein**. Each of the peripheral fibril is a **triplet**. The adjacent triplets are also linked (**AC linker**). The central part of the proximal region of the centriole is also proteinaceous and called the **hub**, which is connected with tubules of the peripheral triplets by **radial spokes made of protein**. The centrioles form the **basal body of cilia or flagella**, and **spindle fibres** that give rise to spindle apparatus during cell division in animal cells.

Question: NEET-2024



Match List-I with List-II.

Column-I

A. Nucleolus ✗

B. Centriole →

C. Leucoplasts →

D. Golgi apparatus →

Column-II

I. Site of formation of glycolipid

II. Organization like the cartwheel

III. Site for active ribosomal RNA synthesis

IV. For storing nutrients

Choose the correct answer from the options given below

1

A-(III), B-(II), C-(IV), D-(I) ✓

2

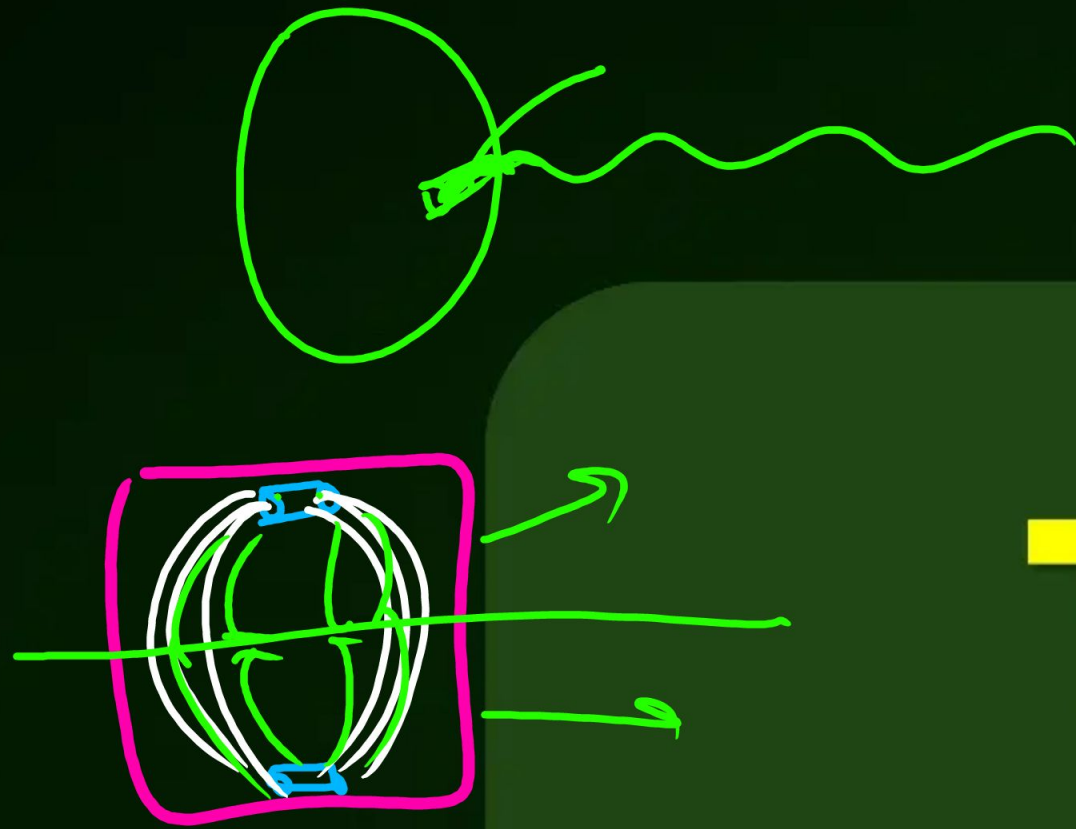
A-(II), B-(III), C-(I), D-(IV) ✗

3

A-(III), B-(IV), C-(II), D-(I) ✗

4

A-(I), B-(II), C-(III), D-(IV) ✗



Thank
me